



SIMATS ENGINEERING
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Technical Presentation
UNIVERSITY ATTENDANCE SYSTEM USING QUEUE
CSA0314 - DATA STRUCTURES

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Problem & Requirement

Problem:

- 1.Attendance is recorded student by student.
- 2.Students are recorded in sequential order.
- 3.New attendance records arrive continuously.
- 4.The system needs efficient insertion of new records.

Requirements:

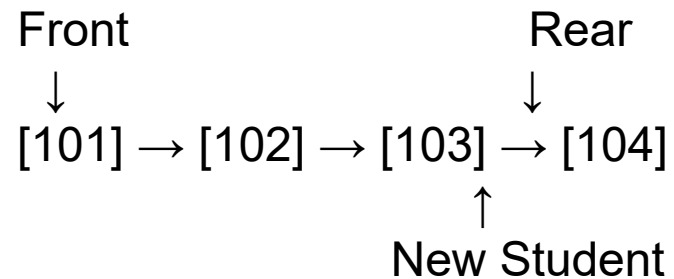
- 1.Store student attendance.
- 2.Maintain sequential order.
- 3.Insert new attendance records.
- 4.Process records in order.



Suitable Data Structure: Graph



- A Queue is a linear data structure.
- It follows FIFO (First In, First Out).
- Front → First attendance record.
- Rear → Latest attendance record.
- New attendance is inserted at the Rear





Insertion Using Enqueue Operation



Steps:

Select the rear position.

Insert the new attendance record.

Move the rear position.

Continue recording the next student.

Before:

$[101] \rightarrow [102] \rightarrow [103]$

New Student: 104

After:

$[101] \rightarrow [102] \rightarrow [103] \rightarrow [104]$



Why Queue is Appropriate?

Data structure Suitability

DATA STRUCTURE	SUITABILITY
ARRAY	MODERATE
LINKED LIST	SUITABLE
STACK	NOT SUITABLE
TREE	LIMITED
GRAPH	NOT SUITABLE
QUEUE	HIGHLY SUITABLE



REASON FOR QUEUE

Reason:

- Maintains sequential order.
- New records are inserted at the rear.
- Follows FIFO.
- Provides efficient insertion.
- Enqueue operation $\rightarrow O(1)$

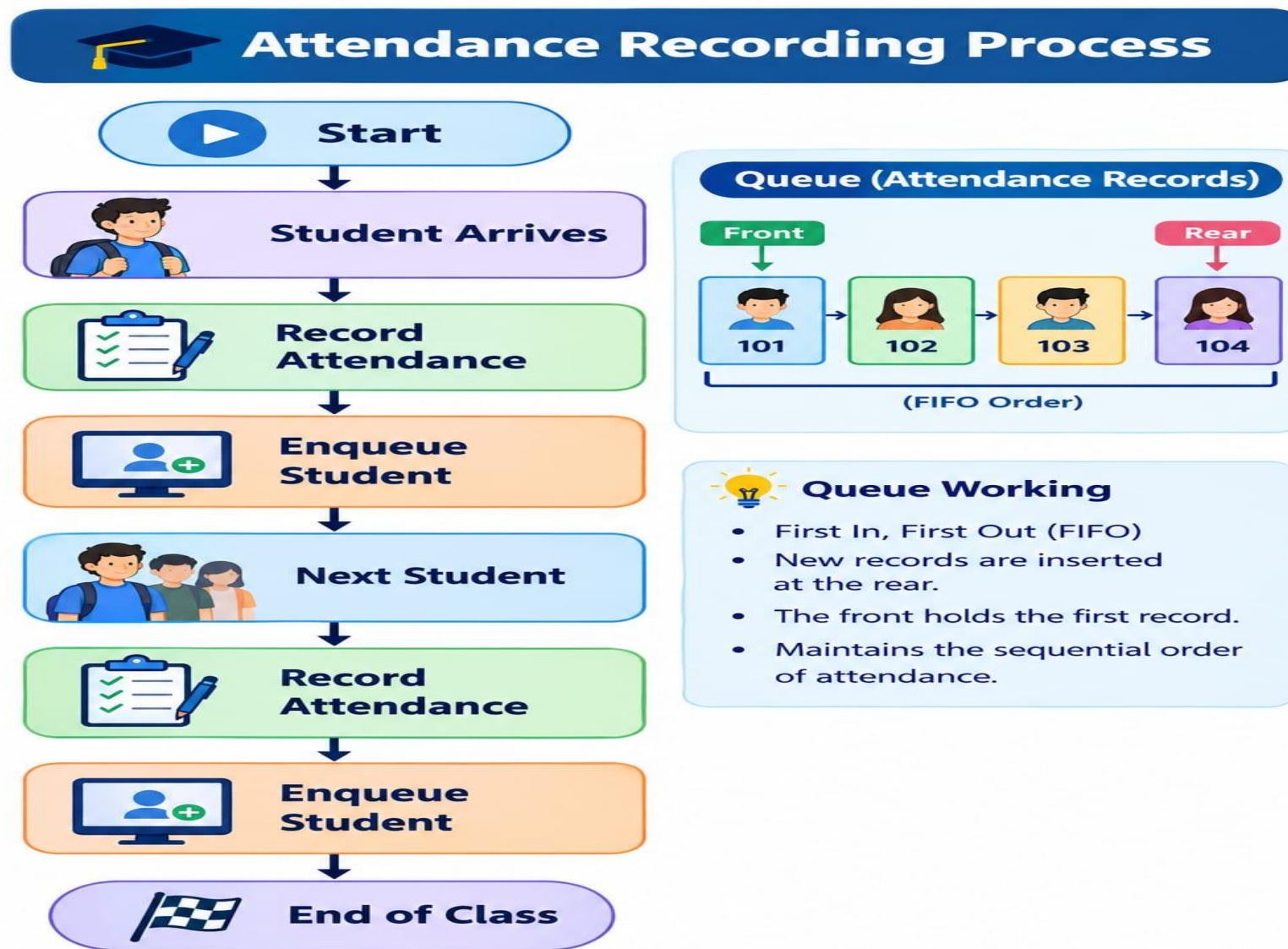


Advantages of Queue

- Maintains student attendance sequence.
- Supports efficient insertion.
- Follows FIFO order.
- Simple to implement.
- New records are added at the rear.
- Suitable for continuous attendance recording.
- Helps process attendance in order.



ATTENDANCE RECORDING PROCESS





Conclusion

- A Queue is suitable for the university attendance system.
- Attendance is recorded sequentially.
- Queue maintains the FIFO order.
- New attendance is inserted at the Rear.
- Enqueue provides $O(1)$ insertion efficiency.
- Queue is simple and efficient for sequential attendance recording.



THANK
YOU